Agent-Based PCB Analysis using Computer Vision and OCR:

Experimental Evaluation and System-Level Insights

1. Introduction

This work presents an agent-based system for analyzing printed circuit board (PCB) images using classical computer vision (CV), optical character recognition (OCR), and large language model (LLM)-based reasoning. The objective is to estimate PCB complexity and approximate the bill of materials (BOM) from a single image.

Real-world PCB images differ significantly from curated datasets. They exhibit high component density, small object sizes, complex textures from traces, and inconsistent text visibility. These factors introduce uncertainty at the perception level, which propagates through downstream reasoning.

Rather than focusing solely on system functionality, this work investigates how variations in image quality and board complexity affect system behavior. The goal is to identify failure modes, explain their underlying mechanisms, and analyze how uncertainty propagates through a multimodal perception–reasoning pipeline.

2. System Architecture

The system is structured as a modular agent-based pipeline combining perception and reasoning components.

The CV module performs grayscale conversion, thresholding, and contour detection to extract geometric features such as region count, area statistics, and spatial coverage. These detected regions are heuristically classified into categories such as ICs, capacitors, and unknown types. The CV pipeline does not detect true electronic components, but rather **contour-based regions derived from pixel intensity gradients**, making it sensitive to image quality and visual structure.

The OCR module (PaddleOCR) extracts textual information such as reference designators (R, C, U, J) and potential IC markings. OCR outputs may include partial or noisy tokens depending on text visibility and resolution.

The agent-based reasoning layer uses an LLM to invoke tools (`run_cv`, `get_ic_info`, `get_component_stats`) and generate outputs including PCB type, complexity, and estimated BOM. A reflection step refines the final response.

This architecture enables flexible reasoning but implicitly assumes that perception outputs are reliable and internally consistent, an assumption that is examined in this work.

3. Experimental Setup

The system was evaluated under controlled variations in input quality using two PCB images:

- **Image 1 (Clean PCB):** Clear component boundaries, moderate density, readable text
- **Image 2 (Dense PCB):** High component density, overlapping structures, limited text visibility

Two perturbation types were applied:

- **Resolution Scaling:** Images resized to simulate reduced resolution (1.0, 0.75, 0.5, 0.25, 0.1)
- **Distance Simulation:** Images downscaled and upsampled to original size, introducing blur and loss of high-frequency detail

Each transformed image was processed through the full pipeline. Multiple runs on identical inputs were also performed to evaluate output stability. Results were recorded in structured logs, including perception outputs, OCR detections, and final reasoning results.

4. Results and Analysis

System behavior is primarily determined by the quality and stability of perception outputs, with the reasoning layer acting as a downstream amplifier rather than a corrective mechanism.

4.1 Effect of PCB Complexity

The system produces significantly different perception outputs for clean and dense PCBs.

The clean PCB yields approximately 350 detected regions with moderate OCR success, while the dense PCB produces approximately 850 detected regions, with over 95% classified as unknown and minimal OCR output.

This discrepancy arises because the CV module measures **contour-based regions rather than physical components**. Dense boards contain overlapping edges, traces, and fine textures that generate additional contours, leading to **over-segmentation** and inflated region counts. In contrast, well-separated structures in clean images produce more stable detections.

As a result, higher detected counts do not necessarily indicate higher physical complexity, but may instead reflect visual clutter and segmentation artifacts.

4.2 Sensitivity to Resolution

Reducing image resolution causes a rapid, non-linear collapse in detected regions. For the clean PCB, detected regions decrease from approximately 350 at full resolution to zero at the lowest scale.

Downscaling disproportionately removes **high-frequency spatial details**, which correspond to small components. As these details disappear, edges become indistinguishable, and contour detection fails.

As contours vanish, the spatial structure of detected regions collapses, effectively removing meaningful structural information required for reasoning. This results in a **topological collapse** of detected structures rather than gradual degradation.

The reasoning layer reflects this loss, with predicted complexity decreasing or becoming undefined as detected region counts approach zero.

4.3 Behavior under Blur and Distance

Distance simulation produces non-intuitive behavior, where detected region counts can increase under blur (e.g., from approximately 850 to over 1000 regions in dense PCB experiments).

Blur redistributes pixel intensity gradients across neighboring regions. After thresholding, these gradients form additional boundaries, which are interpreted as contours. This leads to **systematic over-segmentation** and artificial inflation of region counts.

This inflated region count propagates to the reasoning layer, often resulting in overestimation of PCB complexity and wide or inconsistent BOM estimates. Thus, image degradation can both reduce and inflate detections depending on the type of transformation.

4.4 OCR Dependence on Visual Quality

OCR performance varies significantly across images and conditions.

The clean PCB produces multiple detections of reference markers and partial IC labels, while the dense PCB yields minimal or no usable text. However, OCR degradation is not purely binary. Outputs may include **partial, merged, or incorrect tokens**, resulting in a low signal-to-noise ratio.

These noisy outputs can still influence reasoning, introducing weak or misleading signals. In some cases, OCR suggests IC names that are inconsistent with CV outputs, contributing to conflicting evidence within the system.

Thus, OCR provides uncertain and sometimes unreliable information, especially in dense or low-quality images.

4.5 Limitations of Heuristic Classification

Across both images, the majority of detected regions are classified as “unknown” (approximately 80–95%).

Heuristic classification based on geometric features is insufficient to distinguish visually similar components, particularly at small scales. As a result, the system detects many regions but provides limited semantic understanding.

This weak semantic representation limits the effectiveness of downstream reasoning.

4.6 Instability in Agent-Based Reasoning

Repeated runs on identical inputs produce different outputs, including variations in PCB type and BOM estimates.

While the LLM behaves deterministically for a given input, small variations in perception outputs (e.g., region counts, OCR tokens) alter the input state, leading to different reasoning trajectories. Additionally, inconsistencies between CV and OCR signals introduce conflicting evidence.

For example, CV may indicate a high number of IC-like regions while OCR detects none, leading to contradictory interpretations. The system does not explicitly resolve such conflicts, resulting in variability and instability in final outputs.

4.7 Error Propagation and Lack of Uncertainty Modeling

The system assumes deterministic correctness of perception outputs. No mechanism exists to represent uncertainty or evaluate the reliability of different signals.

As a result, the system produces confident outputs even when input data is incomplete, noisy, or contradictory. For example, high complexity predictions are generated even when OCR fails and component detection is unreliable.

This leads to **systematic overconfidence**, where uncertainty in perception is not reflected in the final output. Errors introduced in perception are propagated and amplified through reasoning.

4.8 Pipeline Robustness

Error handling mechanisms were introduced to convert runtime failures into structured outputs, allowing the system to continue execution and record failure states instead of terminating.

This includes handling OCR failures, invalid outputs, and tool execution errors. However, orchestration-level issues remain, including:

- Invalid JSON outputs from the LLM
- Incorrect or undefined tool calls
- Encoding-related errors

These issues highlight the need for stronger validation and control within agent-based pipelines.

5. Limitations

The system exhibits several limitations:

- Perception is not scale-invariant and is highly sensitive to image degradation
- Detection produces artifacts, including both over-segmentation and under-segmentation
- OCR provides noisy and unreliable signals in complex scenes
- Semantic understanding is limited due to heuristic classification
- Reasoning is sensitive to input variability and lacks stability
- The system does not model or propagate uncertainty
- Validation of intermediate outputs is limited

These limitations collectively reduce reliability in real-world conditions.

6. Proposed Improvements

Improving system performance requires addressing both perception and reasoning.

On the perception side, multi-scale detection and learning-based models (e.g., YOLO) can improve robustness to scale and density. OCR performance can be enhanced through preprocessing, text region detection, and filtering based on expected PCB label patterns.

On the reasoning side, structured validation, strict output constraints, and fallback mechanisms are necessary. Incorporating confidence scores and probabilistic reasoning can enable the system to account for uncertainty and conflicting signals.

A hybrid approach combining classical CV and learned models, along with uncertainty-aware reasoning, is required for reliable performance.

7. Conclusion

This work demonstrates that PCB analysis using CV, OCR, and LLM-based reasoning is feasible but highly sensitive to input quality and scene complexity.

Experimental results show that perception modules are the primary bottleneck. Variations in resolution, blur, and visual density significantly affect detected structures, often in non-intuitive ways such as over-segmentation under blur and collapse under scaling.

The reasoning layer does not correct these errors; instead, it propagates and amplifies upstream uncertainty. In multimodal AI systems, reasoning cannot compensate for unreliable perception, but is constrained by it.

Robust performance therefore requires not only improved perception, but also mechanisms to represent, propagate, and reason under uncertainty.